Effects of the Genetically Reduced Feather Coverage in Naked Neck and Featherless Broilers on Their Performance Under Hot Conditions

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ABSTRACT Under hot conditions, contemporary commercial broilers do not reach their full genetic potential for growth rate, body weight (BW), or breast meat yield because dissipation of their excessively produced internal (metabolic) heat is hindered by the feathers. Therefore, it was hypothesized that heat stress can be alleviated by using the naked-neck gene (*Na*) or the featherless gene (*sc*). The study consisted of 4 experimental genetic groups (fully feathered, heterozygous naked neck, homozygous naked neck, featherless), progeny of the same double-heterozygous parents (*Na/na +/sc*), and commercial broilers. Birds from all 5 groups were brooded together until d 21 when one-half of the birds from each group were moved to hot conditions (constant 35°C), and the others remained under comfortable conditions (constant 25°C). Individual BW was recorded from hatch to slaughter at d 45 and 52 at 25 and 35°C, respectively, when breast meat, rear part, heart, and spleen weights were recorded. Body temperature was recorded weekly from d 14 to 42. Feather coverage significantly affected the thermoregulatory capacity of the broilers under hot conditions. With reduced feather coverage (naked-neck), and more so without any feathers (featherless), the birds at 35°C were able to minimize the elevation in body temperature. Consequently, only the featherless birds exhibited similar growth and BW under the 2 temperature treatments. The naked-neck birds at 35°C showed only a marginal advantage over their fully feathered counterparts, indicating that 20 to 40% reduction in feather coverage provided only limited tolerance to the heat stress imposed by hot conditions. Breast meat yield of the featherless birds was much greater (3.5% of BW, approximately 25% advantage) than that of their partly feathered and fully feathered counterparts and the commercial birds under hot conditions. The high breast meat yield (at both 25 and 35°C) of the featherless broilers suggests that the saved feather-building nutrients and greater oxygen-carrying capacity contribute to their greater breast meat yield. Because of these results, further research on genetically heat-tolerant broilers should focus on the featherless phenotype.

Key words: featherless broiler, naked-neck broiler, heat stress, breast meat yield, body temperature

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INTRODUCTION

Tremendous genetic progress has been achieved in broiler growth rate (**GR**) and meat yield since the 1950s (Havenstein et al., 1994a,b; 2003a,b). However, the elevated genetic potential for rapid growth and the consequent desirable reduction in time to marketing have not been fully expressed under hot conditions (Cahaner, 1996). Greater GR of broilers is driven by greater rate of feed intake and metabolism, and elevated production of internal (metabolic) heat. Hot conditions negatively affect high-GR broilers by hindering dissipation of the excessive internally produced heat, leading to a lethal elevation in body temperature (**BT**). To avoid heat-induced mortality, broilers acclimate to hot conditions by reducing feed intake (Eberhart and Washburn, 1993a,b; Cooper and Washburn, 1998; Deeb and Cahaner, 1999, 2002) resulting in depressed GR, lower final BW, and poorer breast meat yield (Cahaner and Leenstra, 1992; Leenstra and Cahaner, 1992; Cahaner et al., 1995; Settar et al., 1999).

The modern broiler industry continues to expand to hot-climate developing countries where climatic control of broiler houses is lagging because of high installation and operational costs and unreliable supply of electricity. Moreover, in temperate-climate, developed countries, broiler production is negatively affected by heat,

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because broilers are selected for greater GR and BW and consequently generate more heat (Sandercock et al., 1995) and thus need lower ambient temperature (**AT**) to maintain normal BT and express their genetic potential for rapid growth (Emmans and Kyriazakis, 2000). With the limited availability and rising cost of energy, cooling broilers is becoming an economical and political burden on broiler production in developed countries, and therefore breeding for adaptation to heat should become a strategic goal. Commercial local breeding programs under hot conditions have been applied successfully in India (Jain, 2000, 2004). When compared under local hot conditions, imported high-GR broiler stocks have been inferior to the locally bred stock. However, the GR and BW of the locally bred birds are much lower than that of high-GR stocks under comfortable AT, suggesting that conventional broilers cannot be bred to exhibit high GR and high BW under hot conditions. In most hot-climate countries, customers traditionally prefer to buy live broilers with a small body size (~1.5 kg). However, broilers produced for mechanical slaughtering and processing must have a large BW at marketing and a high yield of quality meat—the traits most depressed in high-GR broilers reared under hot conditions (Leenstra and Cahaner, 1992; Mitchell and Sandercock, 1997; Sandercock et al., 2001). Currently, production of carcass parts and deboned meat also increases in hot-climate countries, for export and to supply the increasing local demand for these products. Therefore, the traditional avoidance of the hot climate's negative effects by marketing small-body live broilers must be complemented by alternative strategies.

The rate of sensible ("dry") heat dissipation is determined by the insulation of the feathers. This insulation is advantageous in slow-growing chickens or when broilers are reared under cool conditions (e.g., Leeson and Walsh, 2004). In high-GR broilers under hot conditions, feather coverage negatively affects thermoregulation because it hinders the dissipation of excessive internal heat (Yahav et al., 1998; Deeb and Cahaner, 1999). Therefore, it has been hypothesized that the negative effects of heat can be alleviated by introducing genes that reduce or eliminate feather coverage into the genetic makeup of high-GR broiler stocks (e.g., Somes and Johnson, 1982; Hanzl and Somes, 1983; Horst and Rauen, 1986; Merat, 1986). Reduced feather mass was also expected to contribute to greater meat yield, if the saved feather-building proteins are directed to build more muscle mass, as suggested by Cahaner et al. (1987).

Many studies have been conducted with the naked-neck (*Na*) gene, which is common in rural chicken populations in hot regions. This codominant gene reduces feather coverage by 20% and 40% in heterozygous (*Na/na*) and homozygous (*Na/Na*) chickens, respectively (Crawford, 1976; Cahaner et al., 1993; Yunis and Cahaner, 1999; N'dri et al., 2007). In the 1980s it was suggested that heat tolerance of chickens can be improved by the *Na* gene (Hanzl and Somes, 1983; Horst and Rauen, 1986; Merat, 1986). Under hot conditions, naked-neck broilers exhibit a greater rate of heat dissipation (Yahav et al., 1998) and better thermoregulation (Deeb and Cahaner, 1999), resulting in greater actual GR and meat production than their fully feathered counterparts (Cahaner et al., 1993; Yalcin et al., 1997, 1999; Deeb and Cahaner, 2001). However, in the same studies, the naked-neck broilers raised at 25°C were superior to their naked-neck counterparts under hot conditions, suggesting that the 20 or 40% reduction in feather coverage provides only partial heat tolerance. Therefore, it was hypothesized that complete feather elimination is required to maximize heat tolerance of high-GR broilers (Cahaner et al., 2003; Cahaner and Deeb, 2004).

Abbott and Asmundson (1957) reported on a recessive mutation called scaleless that blocks feather formation in homozygous (*sc/sc*) chickens. This spontaneous mutation was found in the New Hampshire breed, which is characterized by substantially lower GR and BW than those of meat-type chickens; therefore, the featherless mutants were not considered for practical purposes (Somes, 1990). In the late 1970s, experimental featherless broilers were derived from a cross between the scaleless mutant and contemporary broilers. Their GR and carcass composition (Somes and Johnson, 1982) and cooking characteristics (Somes and Wiedenhefft, 1982) were tested. The results suggested an advantage of the featherless birds under hot conditions, but the effects were small because the GR of the birds used in these studies was very low: maximum GR of 30 g/d and average BW of approximately 1,200 g at 8 wk (compared with about 100 g/d in today's broilers that reach the same BW in half the time).

A novel experimental line was developed as a basis for this study, segregating for the *Na* and *sc* genes and with GR greater than that of the birds studied by Somes and Johnson (1982) and Somes and Wiedenhefft (1982), yet substantially inferior to contemporary commercial high-GR broiler stocks. The objective of this study was to compare the actual GR and several performance-related traits of featherless broilers, naked-neck broilers, and normally feathered broilers (all progeny of the same double heterozygous parents) and commercial broilers as an industry reference, under hot (35°C) versus comfortable (25°C) conditions.

MATERIALS AND METHODS

Experimental Stock

The birds used in this study were progeny of intermating among parents that were heterozygous for 2 major genes: naked neck (*Na*) and scaleless (*sc*). The grandparents used to produce these double-heterozygous parents (*Na/na*, *+/sc*) were featherless females (*sc/sc*) and homozygous naked-neck males (*Na/Na*). The naked-neck grandparents were taken from an ex-

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perimental broiler line that was used in several studies during the 1990s (e.g., Cahaner et al., 1993; Yalcin et al., 1997; Deeb and Cahaner, 2001). The genetic potential for GR and meat yield of this line was less than that of contemporary commercial broiler stocks because it had been relaxed for several generations. The featherless grandparents descended from the original New Hampshire scaleless line after only one cycle of backcross to a high-GR broiler stock (Cahaner and Deeb, 2004); therefore, their genetic potential for GR and meat yield was also less than that of contemporary commercial broiler stocks.

Mating among the double-heterozygous *(Na/na +/sc)* parents produced progeny segregating to 3 genotypes in the *Na* gene (*na/na*, *Na/na*, *Na/Na*) as well as in the *sc* gene (*+/+*, *+/sc*, and *sc/sc*). Because only *sc/sc* birds are featherless and this genotype does not allow expression of the *Na* genotypes, there were 4 phenotypic groups of sib progeny: normally feathered (***na/na***), heterozygous naked neck (***Na/na***), homozygous naked neck (***Na/Na***)—all either *+/+* or *+/sc* in the *sc* gene; and featherless (***sc/sc***) that are *na/na* or *Na/na* or *Na/Na* in the *Na* gene. There was also a group of commercial (**Comm**) broilers as an industry reference.

Experimental Design

A total of 261 chicks from all 5 groups were brooded together in a single room on deep litter under standard broiler management. On d 21, half of the birds from each group were moved to a second room and reared together on deep litter under hot conditions (constant 35°C) to 52 d of age. The remaining birds were reared together in the first room under comfortable AT (constant 25°C) to 46 d of age. Relative humidity was around 70% in both rooms. There was only 1 room per AT treatment, but the conditions in these rooms were continuously monitored and equalized (except AT) to minimize the risk of confounding random room differences with the fixed AT effects. The birds in both rooms were checked regularly; there were no cases of skin problems or cannibalism in the featherless birds and in the other groups. Mortality during the trial was very low (0 to 2 birds) in the 4 slow-growing experimental groups, but 10 birds (about 20%) of the Comm birds died, apparently due to heat stress. The final number of birds per group and the AT conditions are presented in Table 1. Except for the different AT, the same management was applied in the 2 rooms, with 23 h/d of light and ad libitum feeding of commercial diets.

Measurements

The BW of each bird was measured 3 times per week during the first 4 wk (on d 0, 3, 5, 7, 10, 12, 14, 17, 19, 21, 24, 26, and 28), twice per week in the following 3 wk (on d 31, 35, 38, and 42), and on d 45 when the trial in the 25°C room was terminated. In the 35°C room, the trial was terminated a week later; hence, BW was measured also on d 49 and 52. For each bird, average daily BW gain (**DBWG**) was calculated for each of the first 5 wk, for the 10 d from 35 to 45 d, and for the last week (45 to 52 d) for the 35°C groups. Birds from all groups were reared together, so feed consumption was not measured.

About 14 birds per group (Table 2) were randomly selected and marked, and their BT was measured at 14, 28, 35, and 42 d of age. Body temperature was measured by inserting the probe of a digital thermometer about 2 cm into the cloaca of the chick for 10 to 15 s, until the reading became stable. Hematocrit was determined for the same birds at 34 d of age.

On d 42, the sex of each bird was determined based on comb size and shank thickness. At the end of the trial (d 46 and 52 in the 25 and 35°C rooms, respectively), about 70 birds per room were randomly selected for carcass measurements (bird numbers per group are presented in Table 3). Feed was removed for 10 h before the birds were killed by cervical dislocation and weighed. The feathers were plucked and the bodies were weighed again to determine the weight of the feathers of each bird. All carcasses, including the featherless ones, were free of downgrades; they were opened and the sex of each bird was confirmed by presence of ovary or testicles. The breast meat (pectoralis major and pectoralis minor) was removed from each carcass (by a single operator) and weighed. The rear part of each carcass, consisting of the pelvis, thighs, and drumsticks (bones included) was weighed. The spleens and hearts were also weighed. After determining the heart's whole weight (total ventricle, **TV**), it was dissected to record the weight of the right ventricle (**RV**), and the RV:TV ratio—an expression of the work load on the right ventricle (Julian, 1993)—was calculated for each bird.

Statistical Analysis

The data of each measurement were subjected to a 3-way ANOVA with the 5 genetic groups, 2 AT treatments (25 vs. 35°C) and sex as main effects, and all interactions. Statistical analyses were conducted using the JMP software (SAS Institute, 2007). There were few significant interactions with sex; hence, means of males and females together are presented in all tables and graphs.

With a substantially greater genetic potential for rapid growth, the commercial broilers (Comm group) were much more affected by hot conditions than the other 4 groups, leading to highly significant interactions between AT and the Comm group vs. the other 4 groups. Therefore, the Comm group was excluded from the ANOVA analyses aimed at determining potential interactions between AT and levels of feather coverage. When this interaction was significant, the AT effect was tested within each group using Student's *t*-test, and the differences between groups were compared using Tukey's honestly significant difference test.

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Table 1. Least squares means of daily BW gain (DBWG, g/d) by week and BW at 45 and 52 d of age, in commercial broilers (Comm) and in the experimental line with 4 segregating phenotypes [normally feathered (*na/na*), heterozygous naked neck (*Na/na*), homozygous naked neck (*Na/Na*), and featherless (*sc/sc*)] reared under comfortable (constant 25°C) and hot (constant 35°C) conditions[1]

	25°C					35°C					*P* (*F*)	Difference[2] [(35°C − 25°C)/25°C, %)]				
Item	Comm	*na/na*	*Na/na*	*Na/Na*	*sc/sc*	Comm	*na/na*	*Na/na*	*Na/Na*	*sc/sc*	G × AT[3]	Comm	*na/na*	*Na/na*	*Na/Na*	*sc/sc*
n	18	23	33	16	35	21	23	33	19	40						
DBWG																
d 0 to 7	14.0^{a}	8.9^{b}	9.0^{b}	8.1^{b}	7.3^{b}	14.9^{a}	8.3^{b}	9.2^{b}	7.9^{b}	8.3^{b}	NS	6.4	−6.8	2.2	−2.5	13.7
d 7 to 14	33.4^{a}	18.9^{b}	18.7^{b}	17.8^{b}	17.6^{b}	36.0^{a}	19.8^{b}	19.7^{b}	18.0^{b}	17.2^{b}	NS	7.8	4.8	5.3	1.1	−2.3
d 14 to 21	55.3^{a}	30.2^{b}	31.2^{b}	30.0^{b}	27.4^{b}	56.7^{a}	29.8^{b}	31.2^{b}	29.1^{b}	27.3^{b}	NS	2.5	−1.3	0.0	−3.0	−0.4
d 21 to 28	66.8^{a}	36.2^{b}	38.5^{b}	37.8^{b}	32.5^{c}	51.5^{a}	30.8^{b}	33.1^{b}	32.3^{b}	33.6^{b}	0.0136	−22.9***	−14.9**	−14.0**	−14.6*	3.4
d 28 to 35	84.5^{a}	43.0^{b}	46.6^{b}	44.2^{b}	39.7^{c}	42.7^{a}	29.6^{c}	33.1^{bc}	33.5^{bc}	36.9^{b}	0.0002	−49.5***	−31.2***	−29.0***	−24.2**	−7.1
d 35 to 45	95.0^{a}	56.8^{b}	57.3^{b}	56.5^{b}	48.4^{c}	28.6^{d}	33.7^{c}	41.0^{b}	42.5^{b}	45.2^{a}	0.0001	−69.9***	−40.7***	−28.4***	−24.8**	−6.6
BW (g)																
d 45	2,770.8^{a}	1,580.8^{b}	1,623.3^{b}	1,569.4^{b}	1,394.3^{c}	1,757.6^{a}	1,234.1^{c}	1,331.6^{bc}	1,314.5^{bc}	1,354.0^{b}	0.0013	−36.6***	−21.9***	−18.0***	−16.2**	−2.9
d 45 to 52	—	—	—	—	—	2.2^{d}	24.8^{c}	28.4^{c}	36.3^{b}	42.3^{a}	—	—	—	—	—	—
d 52	—	—	—	—	—	1,773.0^{a}	1,407.8^{d}	1,530.4^{c}	1,575.6^{c}	1,650.0^{b}	—	—	—	—	—	—

$^{a–d}$Means within age and conditions (25°C or 35°C) without common superscript differ significantly ($P < 0.05$).

[1]Least squares means were calculated from males and females, because there were no significant interactions with sex.

[2]Heat effects are presented as relative (%) differences between corresponding means at 35°C and 25°C, divided by the mean at 25°C.

[3]Significance of the interaction between the 4 segregating genotypes (G; *na/na, Na/na, Na/Na, sc/sc*) and the ambient temperatures (AT; 25 vs. 35°C).

*, **, ***Means under 25°C vs. 35°C, within genetic group and age, differ at $P < 0.05$, $P < 0.01$, and $P < 0.001$ significance level, respectively.

Table 2. Least squares means of body temperature (BT, °C) at 14, 28, 35, and 42 d of age, in commercial broilers (Comm) and in the experimental line with 4 segregating phenotypes [normally feathered (*na/na*), heterozygous naked neck (*Na/na*), homozygous naked neck (*Na/Na*), and featherless (*sc/sc*)] reared under comfortable (constant 25°C) and hot (constant 35°C) conditions[1]

	25°C					35°C					*P* (*F*)	Difference[2] (35°C − 25°C)				
Item	Comm	*na/na*	*Na/na*	*Na/Na*	*sc/sc*	Comm	*na/na*	*Na/na*	*Na/Na*	*sc/sc*	G × AT[3]	Comm	*na/na*	*Na/na*	*Na/Na*	*sc/sc*
n	6	6	7	7	8	7	7	7	8	11	—	—	—	—	—	—
BT: d 14	41.7^{a}	41.7^{a}	41.3^{ab}	41.2^{ab}	41.0^{b}	41.7^{a}	41.8^{a}	41.5^{a}	41.3^{ab}	40.9^{b}	0.1734	0.0	0.1	0.2	0.1	−0.1
BT: d 28	41.4^{a}	41.4^{a}	41.6^{a}	41.6^{a}	41.5^{a}	43.4^{a}	42.2^{b}	41.9^{bc}	41.8^{bc}	41.4^{c}	0.0130	2.0***	0.8**	0.3	0.2	−0.1
BT: d 35	41.7^{a}	41.7^{a}	41.7^{a}	41.4^{a}	41.5^{a}	43.9^{a}	43.1^{ab}	42.8^{ab}	42.5^{b}	41.6^{c}	0.0043	2.2***	1.4***	1.3***	1.1**	0.1
BT: d 42[4]	43.4^{a}	42.5^{b}	42.3^{b}	42.0^{b}	41.6^{c}	43.8^{a}	43.5^{a}	43.1^{b}	42.7^{c}	41.8^{d}	0.0483	0.4	1.2***	0.8**	0.6*	0.2
BT d 42 − BT d 35[4]	1.7	0.8	0.7	0.6	0.1	−0.1	0.4	0.3	0.2	0.2	—	−1.3	−0.4	−0.4	−0.4	+0.1

$^{a–d}$Means within age and conditions (25°C or 35°C) without common superscript differ significantly ($P < 0.05$).

[1]Least squares means were calculated from males and females, because there were no significant interactions with sex.

[2]Heat effects are presented as differences between corresponding means at 35°C and 25°C.

[3]Significance of the interaction between the 4 segregating genotypes (G; *na/na, Na/na, Na/Na, sc/sc*) and ambient temperature (AT; 25 vs. 35°C).

[4]On d 42, the climate control in the 25°C room broke for several hours; hence, AT was about 32°C (instead of 25°C) when BT was measured. To illustrate the effect of this accidental event, the bottom row (BT d 42 − BT d 35) presents the elevation in BT from d 35 to 42 in each group.

*, **, ***Means under 25°C vs. 35°C, within genetic group and age, differ at $P < 0.05$, $P < 0.01$, and $P < 0.001$ significance level, respectively.

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Table 3. Least squares means[1] of BW at slaughter, weight and percentage (of BW) of breast meat, and weight and percentage (of BW) of rear parts, in commercial broilers (Comm) and in the experimental line with 4 segregating phenotypes [normally feathered (*na/na*), heterozygous naked neck (*Na/na*), homozygous naked neck (*Na/Na*), and featherless (*sc/sc*)] reared to 45 d of age under comfortable conditions (constant 25°C) and to 52 d of age under hot conditions (constant 35°C)[1]

	25°C, slaughter at 46 d					35°C, slaughter at 52 d					P(F)	Difference[2] [(35°C − 25°C/25°C, %)]				
Item	Comm	*na/na*	*Na/na*	*Na/Na*	*sc/sc*	Comm	*na/na*	*Na/na*	*Na/Na*	*sc/sc*	G × AT[3]	Comm	*na/na*	*Na/na*	*Na/Na*	*sc/sc*
n	10	8	17	16	21	10	9	16	17	19	—	—	—	—	—	—
BW[4] (g)	2,800.3[a]	1,572.8[b]	1,583.2[b]	1,577.9[b]	1,374.0[c]	1,724.4[a]	1,440.7[c]	1,507.5[b]	1,505.9[b]	1,689.6[a]	0.0002	−38.4***	−8.4*	−4.8	−4.6	23.0**
Breast meat[5] (g)	514.4[a]	231.2[c]	237.0[c]	239.8[c]	273.4[b]	260.1[b]	210.6[b]	222.2[b]	229.2[b]	320.0[a]	0.0551	−49.4***	−8.9	−6.2	−4.4	17.0*
Breast meat (%)	18.7[a]	14.7[c]	15.0[bc]	15.6[b]	18.6[a]	15.1[b]	14.5[c]	14.7[bc]	15.1[b]	18.7[a]	0.0921	−19.3***	−1.4	−2.0	−3.2	0.5
Rear parts[6] (g)	736.8[a]	384.9[c]	416.3[b]	425.2[b]	385.0[c]	503.4[a]	415.4[b]	433.6[b]	437.2[b]	465.5[ab]	0.1845	−31.7***	7.9	4.2	2.8	20.9***
Rear parts (%)	26.8[a]	26.3[a]	26.7[a]	26.8[a]	26.9[a]	29.1[a]	28.9[a]	28.6[a]	28.9[a]	27.5[b]	0.0013	8.6**	9.9**	7.1**	7.8**	2.2

[a–c]Means within age and conditions (25°C or 35°C) without common superscript differ significantly ($P < 0.05$).

[1]Least squares means were calculated from males and females, because there were no significant interactions with sex.

[2]Heat effects are presented as relative (%) differences between corresponding means at 35°C and 25°C, divided by the mean at 25°C.

[3]Significance of the interaction between the 4 segregating genotypes (G; *na/na, Na/na, Na/Na, sc/sc*) and ambient temperature (AT; 25 vs. 35°C).

[4]BW after 10 h of feed withdrawal.

[5]Pectoralis major and pectoralis minor.

[6]Pelvic, thighs, and drumsticks.

*, **, ***Means under 25°C vs. 35°C, within genetic group and age, differ at $P < 0.05$, $P < 0.01$, and $P < 0.001$ significance level, respectively.

RESULTS AND DISCUSSION

Variation in Feather Coverage

The relative weight of the feathers (% of BW) was considered as an expression of feather coverage, with the coverage of the featherless broilers being 0%. Mean feather coverage of the 3 genotypes of the naked-neck gene reflected a "dose effect" of the *Na* allele: 5.37% in the fully feathered (*na/na*), 4.19% in the heterozygous naked neck (*Na/na*), and 3.36% in the homozygous naked neck (*Na/Na*). The feather coverage of the *na/na* birds is in agreement with literature (reviewed by Leeson and Walsh, 2004) reporting feather weight in broilers at marketing age as being between 5 and 6% of BW. Considering the value of 5.37% as full feather coverage (100%), the relative feather coverage of the other 2 genotypes were 78% (4.19/5.37) and 63% (3.36/5.37). These figures are in agreement with previous reports (e.g., Cahaner et al., 1993; N'dri et al., 2007), determining approximately 80 and 60% feather coverage in heterozygous and homozygous naked-neck broilers, respectively.

The effects of feather coverage on broilers were evaluated by comparing 4 distinctive levels: fully feathered (*na/na*), heterozygous (*Na/na*), and homozygous (*Na/Na*) naked neck, and featherless. A special experimental line was established by crossing parents heterozygous for 2 independent major genes, *Na* and *sc*. Thus, the progeny of the same parents segregated to 4 different feather-coverage phenotypes but shared the same average genetic background with regard to all other traits, including those reported in this study. This experimental design allowed determination of the net effects of these 4 genetically induced levels of feather coverage. However, the genetic background of the experimental groups was known to be inferior in performance traits to contemporary commercial broilers; hence, a group of the latter (Comm) was included in the study. This group allowed us to determine the effects of genetic potential for high GR on the response of fully feathered broilers to hot conditions, and to extrapolate on the effects of reduction or elimination of feather coverage on contemporary fast-growing broilers.

GR and BW

The graphs of the growth curves clearly show the greater GR of the Comm broilers compared with the 4 groups in the experimental population segregating for the *Na* and *sc* genes (Figure 1). In the 25°C groups, after 1 wk, the mean BW of the Comm chicks was substantially greater than the means of the other 4 groups, with the advantage increasing to more than 1,000 g at the trial's end, on d 45. At 35°C, the Comm broilers also exhibited greater GR, but only to the fourth week; from d 28, the GR of the Comm broilers was declining, and, on average, their BW did not increase during the extra rearing week (45 to 52 d) of the birds at 35°C.

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Relative to Comm birds, the differences in GR among the 4 segregating genotypes from the experimental line (sharing the same genetic background and differing only in feather coverage) were quite small and not clearly apparent from Figure 1. The DBWG was calculated for each week and presented in Table 1, along with mean BW on d 45 (slaughter age for the 25°C group) and d 52 (slaughter age for the 35°C group). The table also presents the significance of the interaction between AT and the 4 segregating genotypes, and the AT effect on each of the 5 groups. During the 3 brooding weeks there was no significant difference (within group) between the chicks that were later moved to 35°C vs. those that stayed at 25°C (Table 1). The Comm group already exhibited its greater GR from wk 1, with mean DBWG about 70% greater compared with the fully feathered experimental genotype (*na/na*). The 4 genotypes segregating for feather coverage did not differ significantly in DBWG during the brooding period, but in wk 3, the featherless broilers (*sc/sc*) gained slightly less weight than their fully feathered and naked-neck counterparts.

The GR of the Comm broilers at 25°C kept accelerating from 66.8 g/d in wk 4 to 95 g/d during the last 10 d, reaching a mean BW of 2,770 g on d 45. Compared with groups at 25°C, the mean DBWG of the Comm broilers at 35°C was depressed by about 23% in wk 4, by 50% in wk 5, and by 70% during the period from d 35 to 45, to a mean BW of 1,757.6 g, 36.6% less than mean BW of their counterparts at 25°C. From d 45 to 52, the Comm broilers at 35°C hardly gained any weight (mean DBWG = 2.2 g), as evident from their growth curve. This dramatic heat effect on the Comm broilers, in agreement with similar results from previous studies under controlled 35°C (e.g., Deeb and Cahaner, 2002) and undera natural hot climate (e.g., Settar et al., 1999), demonstrates the substantial susceptibility of contemporary high-GR broiler stocks to hot conditions.

At 25°C, the fully feathered birds (*na/na*) and their heterozygous (*Na/na*) and homozygous (*Na/Na*) naked-neck counterparts exhibited similar GR during the 3 periods (d 21 to 28, d 28 to 35, and d 35 to 45), with mean BW of about 1,600 g on d 45. The means of DBWG from d 21 to 45 and BW on d 45 of the featherless birds (*sc/sc*) at 25°C were significantly lower than the corresponding means of their fully feathered and naked-neck counterparts. Apparently, AT of 25°C was below the optimum for the featherless chicks, and they dissipated more energy than they could compensate for by increasing feed consumption.

By wk 4, the 35°C group showed significantly reduced (by about 14%) mean DBWG in the fully feathered and naked-neck birds (*na/na*, *Na/na*, and *Na/Na*), but not in the featherless birds (*sc/sc*). Moreover, in the subsequent periods (d 28 to 35 and d 35 to 45), means of DBWG in the featherless birds at 35°C were similar to the corresponding means in birds at 25°C. In addi-

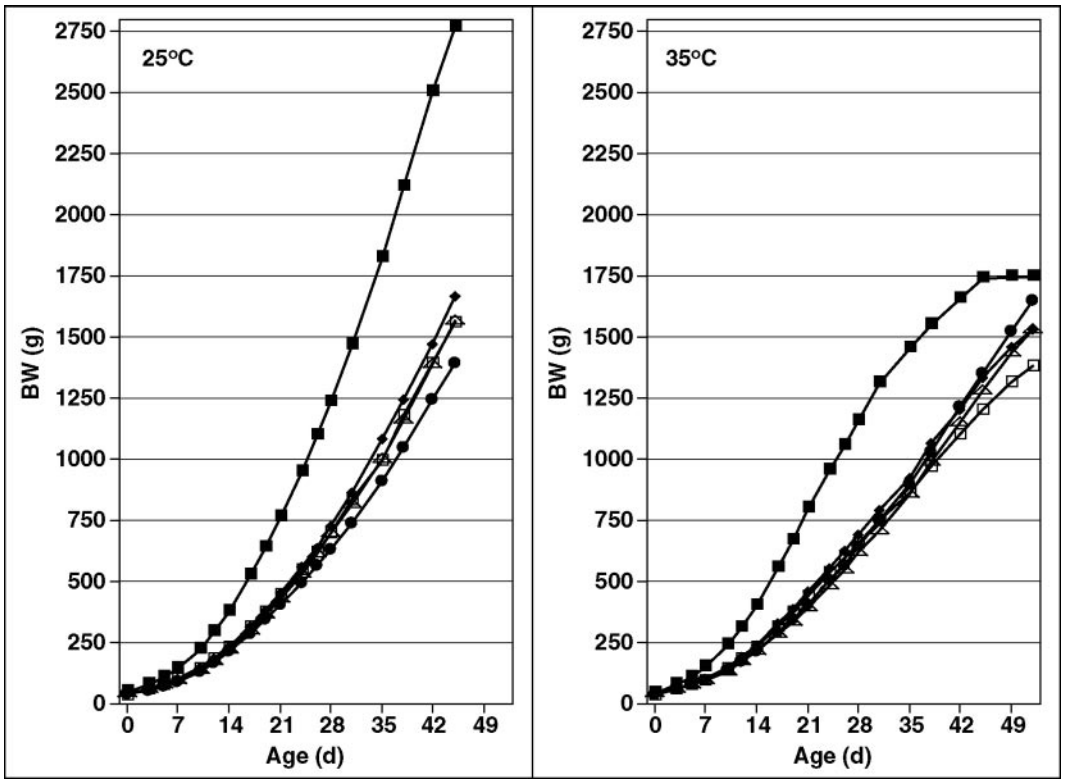


Figure 1. Growth curves (female and male means) to 45 d of age under comfortable conditions (constant 25°C), and to 52 d of age under hot conditions (constant 35°C), of the commercial broilers (Comm, ■) and the experimental line with 4 segregating phenotypes: normally feathered (*na/na*, □), heterozygous naked neck (*Na/na*, ◊), homozygous naked neck (*Na/Na*, Δ), and featherless (*sc/sc*, ●).

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tion, the featherless group at 35°C exhibited the greatest mean DBWG during the last 17 d. Consequently, mean final BW (on d 52) of the featherless broilers was significantly greater than those of their fully feathered (*na/na*) and naked-neck (*Na/na* and *Na/Na*) counterparts. The GR and BW of the 3 fully feathered and naked-neck genotypes at 35°C were similar, but only to 28 d of age. By wk 5, the 2 naked-neck groups gained more BW than their fully feathered counterparts (approximately 33.3 vs. 29.6 g/d). During the period from d 35 to 45, mean DBWG of the 2 naked-neck groups (about 42 g/d) was significantly greater than that of the fully feathered birds (33.7 g/d). Consequently, the heat effect on DBWG during this period was more significant in the fully feathered (*na/na*) group (reduction of 40.7%) than in the 2 naked-neck groups (reductions of 28.4 and 24.8%, respectively). The DBWG of the homozygous naked-neck birds at 35°C was significantly greater than that of their *Na/na* counterparts (36.3 vs. 28.4 g/d) only in the last week. This ranking of mean GR and BW at 35°C reflected the differences in feather coverage between these 3 *Na* genotypes, and accordingly, the featherless birds exhibited the greatest GR and BW among the 4 segregating groups. Similar ranking of the 3 *Na* genotypes at 35°C was found previously (Cahaner et al., 1993), but the present study is the first to show that this gradual effect of reduced feather coverage continues in featherless broilers.

Body Temperature

Body temperature was measured in approximately 7 birds per group and AT combination (Table 2). On d 14, all the birds were together under standard brooding conditions, and accordingly there were no differences in BT between the birds that later were moved to 35°C vs. those that remained at 25°C. It appears that the gradual reduction in feather coverage, from fully feathered (*na/na*), through heterozygous naked-neck (*Na/na*) to homozygous naked neck (*Na/Na*), and then to featherless (*sc/sc*) led to a corresponding reduction in BT on d 14, from 41.8 (*na/na*) to 41.4 (*Na/na*), 41.2 (*Na/Na*), and 40.9°C (*sc/sc*). These differences suggest that under the relatively low AT (about 28°C) at this stage of brooding, and with small body (mean BW on d 14 was about 230 g) and limited appetite of the chicks from the segregating experimental population, those with reduced plumage could not fully counteract the loss of body heat. Indeed, the ranking of these 4 genotypes by their mean BT on d 14 was the same as their ranking according to mean BW on d 21 (end of brooding period), from 455 g (*na/na*) to 403 g (*sc/sc*), with the naked-neck genotypes exhibiting intermediate means (444 and 419 g). However, after an additional 2 and 3 wk at 25°C, the featherless and naked-neck birds did not differ from the fully feathered ones (*na/na* and Comm) in mean BT on d 28 and 35. Apparently, the feed intake capacity and BW of the featherless broilers during wk 4 and 5 were sufficient to maintain normal BT.

Mean BT on d 28 of the fast-growing Comm broilers reared at 35°C was elevated to 43.4°C, 2°C above that of their counterparts reared at 25°C. Mean BT on d 28 of the fully feathered *na/na* group was lower than that of the fully feathered Comm group, and only 0.8°C greater than that of the *na/na* birds reared under the cooler conditions. The different heat effect on BT in the 2 fully feathered groups is explained by the association between BT and BW. Compared with the *na/na* birds, the Comm birds at 35°C grew faster during wk 4 (d 21 to 28, Table 1) and were considerably heavier on d 28 (~1,150 vs. 650 g, Figure 1). Internal heat production is greater in broilers with greater GR (Sandercock et al., 1995); therefore, the Comm birds apparently could not dissipate the excess of internal heat, and hence, the greater elevation in their BT. It is suggested that the elevated BT of the Comm birds at 35°C led to reduced feed intake and GR (Table 1), in agreement with previous reports (Cooper and Washburn, 1998; Deeb and Cahaner, 1999).

The BT on d 28 of the groups with reduced feather coverage (*Na/na*, *Na/Na*, and *sc/sc*) was not elevated in the 35°C group. However, with the greater DBWG (Table 1) and BW (Figure 1) from d 28 to 35, BT on d 35 was greater in the birds reared at 35°C compared with their counterparts at 25°C in all groups except the featherless birds. The gradual reduction in feather coverage of the 4 segregating groups (*na/na*, *na/na*, *Na/Na*, and *sc/sc*) was clearly reflected in the ranking of their mean BT at 35°C on d 42: 43.5, 43.1, 42.7, and 41.8°C, respectively (Table 2). Mean BT of the 3 *Na* genotypes at 35°C ranked similarly in several other studies (Eberhart and Washburn, 1993b; Deeb and Cahaner, 1999; N'dri et al., 2007). Thus, it can be generalized that, with reduced feather coverage, BT is less elevated under hot conditions. Accordingly, with no feathers, the featherless broilers (*sc/sc*) were the only ones to maintain normal BT when reared at 35°C. It appears that because of this capacity of the featherless broilers, they exhibited the greatest GR from d 35 to 52 at 35°C (Table 1).

On d 42, the climate-control system in the 25°C room failed for the whole day, and the AT was about 32°C when BT was measured. This accidental heat stress elevated the d 42 BT, compared with the d 35 BT, in a manner that reflected BW and feather coverage. The greatest BT (43.4°C), 1.7°C above their mean BT a week earlier, was measured on the heavy fully feathered Comm broilers (Table 2); 5 birds (about 20%) in that group died because of this elevation in BT. Mean BT in the fully feathered and naked-neck groups (*na/na*, *Na/na*, and *Na/Na*) were less elevated due to this accidental short-term heat stress in the 25°C room; they were ranked according to their feather coverage: 42.5, 42.3, and 42.0°C, respectively. The featherless birds tolerated this heat stress with mean BT of 41.6°C on d 42, only 0.1°C above their mean BT on d 35.

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Meat Yield

At the end of the trial (d 45 for the 25°C group and d 52 for the 35°C group), meat yield was determined for about one-half of the birds in each group. Table 3 presents the exact number of birds per group and their mean BW upon slaughter after 10 h of feed withdrawal. These means were similar to the corresponding means of the entire groups at these ages (Table 1), indicating that meat yield was measured on representative birds from each genetic group and AT treatment. Aiming to separate the net effect of heat on meat yield from its effect on BW, the birds at 35°C were reared for 1 more week (to d 52), because mean BW on d 45 (Table 1) of all groups (except featherless) were lower than that of their counterparts at 25°C. With the extra week, mean BW of the 52-d-old slaughtered birds from the 2 naked-neck groups at 35°C was only about 5% lower than that of their 45-d-old counterparts in the 25°C group (Table 3). In the fully feathered birds, this difference was about 9% in the *na/na* group and much greater in the Comm group, where mean BW (1,724.4 g) of birds in the 35°C group was 38% lower than that of their counterparts reared to d 45 at 25°C (2,800.3 g). The difference in mean BW between the 2 AT treatments was reversed in the featherless groups of slaughtered birds; it was 1,374 g on d 45 at 25°C, and 1,689.6 g (greater by 23%) on d 52 at 35°C.

The high breast meat yield (**BR%**) of Comm broilers at 25°C (18.7% of BW, Table 3) agrees with recent studies with commercial broilers (e.g., Havenstein et al., 2003b) and reflects the intensive commercial breeding for this trait since early 1990s. In contrast, the experimental line segregating for the *Na* and *sc* genes was derived from a cross between a line of naked-neck broilers that had never been selected for greater BR%, and featherless birds after only 1 backcross of the original New Hampshire scaleless line to a contemporary broiler stock in the early 2000s. Hence, the fully feathered *na/na* group exhibited lower BR% (14.7 and 14.5% at 25 and 35°C, respectively), which is similar to that of commercial broilers of the 1980s and early 1990s (Cahaner and Nitsan, 1985; Leenstra and Cahaner, 1992; Havenstein et al., 1994b). The relatively small reduction in feather coverage slightly increased BR% in the *Na/na* and *Na/Na* naked-neck groups, respectively, to 15 and 14.7% at 25°C, and to 14.7 and 15.1% at 35°C, but only the effect of the *Na/Na* genotype was significant (Table 3), in agreement with 3 previous studies (Cahaner et al., 1993; Deeb and Cahaner, 1999; N'dri et al., 2007). In several other studies, fully feathered birds were compared only with heterozygous naked neck, and, as in the present study, the latter had greater BR%, but the difference was too small to be significant (Hanzl and Somes, 1983; Yalcin et al., 1997, 1999; Yunis and Cahaner, 1999; Deeb and Cahaner, 2001). The results of these studies indicate that the reduction in feather mass in the *Na/na* and *Na/Na* genotypes (Crawford, 1976) elevates BR% independent of AT, possibly because the nutrients that otherwise would have been used to build feathers were deposited in the breast muscle and increased BR%. A possible trade-off between lower feather mass and greater breast meat yield was suggested by Cahaner et al. (1987).

This hypothesis was supported by the featherless broilers; their mean BR% in the 2 AT treatments—18.6 and 18.7% of BW (Table 3)—was much greater than the approximately 15% in their fully feathered and naked-neck partners that share the same genetic background, and similar to the 18.7% in the Comm group at 25°C. Thus, the effect of being featherless, due to the major gene genotype *sc/sc*, was equivalent to the gradual progress in BR% of contemporary commercial broilers following 2 decades of selection applied by breeding companies (Havenstein et al., 2003b). The favorable effect on BR% of being featherless was accentuated at 35°C. As was shown in previous studies (Leenstra and Cahaner, 1992; Cahaner at al., 1993; Deeb and Cahaner, 2001, 2002), broilers with genetic potential for rapid growth and high meat yield are more susceptible to heat. Those studies suggested that to minimize internal heat production when reared under hot conditions, such broilers reduce overall feed intake and growth, with maximal reduction in the growth of the nonfunctional breast muscle. Indeed, BR% of the Comm broilers at 35°C was only 15.1% of BW (Table 3). In absolute terms, mean breast meat weight of Comm birds was 514 g on d 45 at 25°C, and only 260 g on d 52 at 35°C, a reduction of about 50%. In contrast, mean breast meat weight of the featherless birds increased from 273 g at 25°C to 320 g at 35°C, about 100 g greater than their fully feathered and naked-neck counterparts, and 60 g greater than in the Comm broilers.

Means of rear-part weight of the 5 groups at 25°C reflected their BW means; hence, the groups did not differ in the rear weight relative to BW (Table 3). This similarity was expected because the rear (consisting of pelvic and leg bones and muscles) is a functional part of the body, in contrast to the breast muscle. The 4 groups with feathers reared at 35°C were similar in rear-weight percentage, in agreement with N'dri et al. (2007). Mean rear-weight percentage of the featherless birds at 35°C (27.5%) was less than that of their feathered counterparts at 35°C and similar to their featherless counterparts at 25°C (26.9%), being related to the fact that only the featherless broilers had the same BR% under the 2 AT treatments.

Hematocrit, Heart, and Spleen

Hematocrit, heart, and spleen were measured to understand physiological consequences of heat effects on broilers differing in feather coverage, GR, and BW. Hematocrit (Table 4) and BT (Table 2) were measured on the same birds; hearts and spleens were taken from the birds that were slaughtered to measure meat yield (Table 3). At 25°C, mean hematocrit was similar (25.7 and 25.8%) in the 2 groups with full feather coverage,

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whereas greater means (26.5, 28.4, and 30.1%) were exhibited by the *Na/na, Na/Na*, and *sc/sc* groups. It appears that these higher hematocrit levels reflected the increasing coldness felt by the birds in these groups because of their reduced feather coverage. The negative association between coldness and hematocrit has been shown in many studies (e.g., Yahav et al., 1997). It should be noted that mean BT of these birds 1 d later (d 35, Table 2) was similar in all 5 groups at 25°C. Thus, the featherless and naked-neck birds managed to maintain normal BT at 25°C, probably by increasing their metabolism, which required higher oxygen supply; hence, the elevation in hematocrit. The increasing metabolism also required more dietary energy, and because the featherless broilers in this study (as their *na/na, Na/na* and *Na/Na* counterparts) lacked the feed intake capacity of contemporary broilers, the extra use of feed for thermoregulation might explain their lower BW on d 45: 1,394 g compared with 1,580 g reached by their fully feathered counterparts (Table 1). At 35°C, mean hematocrit values of the 5 groups were ranked in reverse order to their BT means (Table 2). Broilers under heat stress reduce their metabolism to minimize internal heat production, and consequently their hematocrit is reduced (Yahav et al., 1997). The featherless birds, and, to a lesser extent, the homozygous naked-neck birds, were the only groups maintaining normal levels of hematocrit when reared at 35°C, similar to that of the feathered groups (Comm and *na/na*) reared at 25°C. These results indicate that without feathers, and, to a lesser extent, with about 40% reduction in feather mass, the 5-wk-old birds at 35°C in this study were not under heat stress.

The hearts were removed from all the slaughtered birds, and their mean weights as a percentage of BW (**HW%**) are shown in Table 4. In a study by Yahav et al. (1997), HW% was negatively correlated with AT in feathered broilers, with BW ranging between 2 and 3 kg; means were 0.53, 0.49, 0.43, 0.42, 0.35, and 0.30% in broilers reared at 10, 15, 20, 25, 30, and 35°C, respectively. That study suggested that HW% indicates the actual AT felt by the broilers: around 0.4% indicates comfortable AT and lesser or greater values indicate hot or cold conditions, respectively. According to this scale, the fully feathered Comm and *na/na* birds and the *Na/na* naked-neck birds were comfortable at 25°C, with mean HW% of 0.38, 0.40 and 0.42%, respectively. The *Na/Na* naked-neck birds (mean HW% = 0.45%) probably felt as if they were under somewhat lower AT. The *sc/sc* birds, without any feathers and with low BW and limited appetite, probably felt as if they were under low AT, as indicated by their mean HW% (0.54%) and hematocrit (30.1%), both the highest of all groups.

The 5 groups at 35°C also varied in HW%; the featherless birds appeared to feel comfortable with a mean HW% of 0.37%, similar to the Comm birds at 25°C (0.38%). The birds in the other groups were stressed by 35°C, with HW% decreasing (0.30, 0.28, 0.26, and

Table 4. Least squares means of hematocrit at 34 d of age, spleen and heart weights (relative to BW), and of the right ventricle to total ventricle weight ratio (RV:TV) in slaughtered commercial broilers (Comm) and in the experimental line with 4 segregating phenotypes [normally feathered (*na/na*), heterozygous naked neck (*Na/na*), homozygous naked neck (*Na/Na*), and featherless (*sc/sc*)] reared to 45 d of age under comfortable conditions (constant 25°C) and to 52 d of age under hot conditions (constant 35°C)[1]

	25°C, slaughter at 46 d				35°C, slaughter at 52 d				P (F)	Difference[3] [(35°C − 25°C/25°C, %)]			
Item[2]	Comm	*na/na*	*Na/na*	*Na/Na*	*sc/sc*	Comm	*na/na*	*Na/na*	*Na/Na*	*sc/sc*	G × AT[4]	Comm	*na/na*	*Na/na*	*Na/Na*	*sc/sc*
Hematocrit (%)	25.7^{b}	25.8^{b}	26.5^{b}	28.4^{ab}	30.1^{a}	22.0^{d}	23.2^{c}	24.3^{bc}	25.3^{ab}	26.0^{a}	0.0350	−14.4**	−10.1*	−8.3*	−10.9*	−13.6**
Heart (% of BW)	0.38^{c}	0.40^{c}	0.42^{bc}	0.45^{b}	0.54^{a}	0.25^{d}	0.26^{d}	0.28^{c}	0.30^{b}	0.37^{a}	0.0001	−34.2***	−35.0***	−33.3***	−33.3***	−31.5***
RV:TV (g/g)	0.15^{b}	0.16^{a}	0.17^{ab}	0.18^{a}	0.19^{a}	0.15^{b}	0.15^{b}	0.16^{ab}	0.17^{a}	0.17^{a}	NS	0.0	−6.2	−5.9	−5.6	−10.5
Spleen (% of BW)	0.11^{a}	0.16^{a}	0.16^{a}	0.17^{a}	0.19^{a}	0.04^{c}	0.05^{c}	0.09^{b}	0.10^{b}	0.16^{a}	0.0007	−63.6***	−68.8***	−43.8***	−41.2***	−15.8**

[a-d]Means within age and conditions (25°C or 35°C) without common superscript differ significantly ($P < 0.05$).

[1]Least squares means were calculated from males and females, because there were no significant interactions with sex.

[2]Number of birds per group (n) as in Table 2 or Table 3 (for hematocrit).

[3]Heat effects are presented as relative (%) differences between corresponding means at 35°C and 25°C, divided by the mean at 25°C.

[4]Significance of the interaction between the 4 segregating genotypes (G; *na/na, Na/na, Na/Na, sc/sc*) and ambient temperatures (AT; 25 vs. 35°C).

*, **, ***Means under 25°C vs. 35°C, within genetic group and age, differ at $P < 0.05$, $P < 0.01$, and $P < 0.001$ significance level, respectively

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0.25%) as feather coverage increased to 60% (*Na/Na*), 80% (*Na/na*), and 100% (*na/na* and Comm), respectively. Thus, the 10 combinations of 2 AT (25 and 35°C) and 5 genetic groups differing in feather coverage and actual GR, resulted in a range of HW% values similar to those obtained by rearing fully feathered broilers under 6 levels of AT from 10 to 35°C (Yahav et al., 1997). There was a highly significant correlation (0.956) between the means of HW% and hematocrit in the 10 groups. Apparently, the level of hematocrit also reflects the actual AT that is felt by the birds; a similar correlation (0.967) between hematocrit and heart weight (as % of BW) was reported also by Yahav et al. (1997).

After determining the heart's whole weight (TV), it was dissected to record the weight of the RV, and the RV:TV ratio was calculated for each bird. This ratio, an expression of right ventricle hypertrophy, is widely used as an indicator of broilers that develop ascites syndrome due to failure to match elevated oxygen demand, often induced by low AT (Julian, 1993). Indeed, group means of RV:TV (Table 4) were greater as the birds had greater reduction in feather coverage, and this trend was more apparent at 25°C than at 35°C. However, all RV:TV means, and even individual RV:TV values of the featherless birds, were <0.25, which is considered as the threshold between healthy birds and those developing ascites. Thus, being without feathers did not induce ascites among the featherless broilers under both AT treatments in this study.

Hot conditions are known to affect the development of immunocompetence in broilers; hence, following Bartlett and Smith (2003), the weight of the spleen (an immune system organ) was determined in all slaughtered birds. In agreement with this cited report, in all 5 groups of the present study, mean spleen weight as percentage of BW (**SW%**) was significantly less under 35°C than under 25°C (Table 4). However, the magnitude of this heat effect appeared to be related to feather coverage. In the 2 fully feathered groups (Comm and *na/na*), SW% was reduced by about 65%; that is, at 35°C the spleen was about one-third of its size at 25°C. In the 2 naked-neck groups (*Na/na* and *Na/Na*) at 35°C, SW% was reduced by about 42% and in the featherless birds (*sc/sc*), SW% was reduced by only 16%. Consequently, mean SW% of the featherless broilers at 35°C was about 4 times greater than those of the fully feathered groups (0.16 vs. 0.04 and 0.05%, Table 4), and also substantially greater than the means of the naked-neck groups (0.09 and 0.10%). These results suggest that featherless broilers, because of their heat tolerance, show superior immunocompetence at 35°C than their counterparts with feathers.

Summary

The amount of feather coverage clearly affected the thermoregulatory capacity of broilers under hot conditions. With reduced feather coverage, and more so without any feathers, the birds under hot conditions were able to maintain normal levels of BT, hematocrit, and HW%. Consequently, the featherless birds exhibited similar GR and final BW under the 2 AT treatments, whereas the Comm broilers exhibited the greatest and earliest depression in GR under hot vs. comfortable temperature. As feather coverage was reduced from the fully feathered *na/na* birds to the homozygous *Na/Na* birds, BR% increased, but the greatest BR% was exhibited by the featherless birds. It appears that in addition to heat tolerance, there are other factors contributing to the greater BR% of the featherless broilers; for example, larger hearts and higher hematocrit (superior oxygen-carrying capacity), and possibly the availability of extra amino acids and energy that are used by standard broilers to build their feathers. Heat tolerance and greater BR% of featherless broilers were suggested by the results of Somes and Johnson (1982) but the effects were small, reflecting the genetically low GR of the experimental birds in that study. In the present study, potential GR and BW of the featherless birds and their fully feathered counterparts were greater; hence, the benefits of being featherless were substantially larger. With no skin or other welfare problems in the featherless broilers and their superior performance under hot conditions, the present study suggests that introduction of the *sc* gene and the featherless phenotype into high-GR broiler stocks is a promising approach to facilitate low-cost production of broiler meat under hot conditions. However, the potential GR and BR% of the featherless birds in this study were still lower than those of contemporary commercial broilers. Therefore, further studies on the effects of the featherless phenotype on growth, feed conversion, meat yield and quality, and welfare, are being conducted using featherless birds with the genetic background of contemporary fast-growing commercial broilers.

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